Name
Date
Course

Limits (Part II)

Finding Limits at Removable Discontinuities (cont.)

Using the Fundamental Principle of Fractions

Find the value of each limit.

① $\lim\limits_{x\to 4} \dfrac{\sqrt{x-3}-1}{x-4} = \lim\limits_{x\to 4}\left[\dfrac{(\sqrt{x-3}-1)}{(x-4)}\dfrac{(\sqrt{x-3}+1)}{(\sqrt{x-3}+1)}\right]$

$= \lim\limits_{x\to 4}\left[\dfrac{(\sqrt{x-3})^2+\sqrt{x-3}-\sqrt{x-3}-1}{(x-4)(\sqrt{x-3}+1)}\right]$

$= \lim\limits_{x\to 4}\left[\dfrac{x-3-1}{(x-4)(\sqrt{x-3}+1)}\right]$

$= \lim\limits_{x\to 4}\left[\dfrac{x-4}{(x-4)(\sqrt{x-3}+1)}\right]$

$= \lim\limits_{x\to 4} \dfrac{1}{\sqrt{x-3}+1}$

$= \dfrac{1}{\sqrt{4-3}+1}$

$= \dfrac{1}{\sqrt{1}+1}$

$= \dfrac{1}{1+1} = \boxed{\dfrac{1}{2}}$

② $\lim\limits_{x\to 64} \dfrac{x-64}{\sqrt{x}-8} = \lim\limits_{x\to 64}\left[\dfrac{(x-64)}{(\sqrt{x}-8)}\dfrac{(\sqrt{x}+8)}{(\sqrt{x}+8)}\right]$

$= \lim\limits_{x\to 64}\left[\dfrac{(x-64)(\sqrt{x}+8)}{(\sqrt{x})^2+8\sqrt{x}-8\sqrt{x}-64}\right]$

$= \lim\limits_{x\to 64}\left[\dfrac{(x-64)(\sqrt{x}+8)}{x-64}\right]$

$= \lim\limits_{x\to 64}(\sqrt{x}+8)$

$= \sqrt{64}+8$

$= 8+8$

$= \boxed{16}$

③ $\lim\limits_{x\to 0} \dfrac{\sqrt{x+49}-7}{x} = \lim\limits_{x\to 0}\left[\dfrac{(\sqrt{x+49}-7)}{x}\dfrac{(\sqrt{x+49}+7)}{(\sqrt{x+49}+7)}\right]$

$= \lim\limits_{x\to 0}\left[\dfrac{(\sqrt{x+49})^2+7\sqrt{x+49}-7\sqrt{x+49}-49}{x(\sqrt{x+49}+7)}\right]$

$= \lim\limits_{x\to 0}\left[\dfrac{x+49-49}{x(\sqrt{x+49}+7)}\right]$

$= \lim\limits_{x\to 0}\left[\dfrac{x}{x(\sqrt{x+49}+7)}\right]$

$= \lim\limits_{x\to 0}\dfrac{1}{\sqrt{x+49}+7}$

$= \dfrac{1}{\sqrt{0+49}+7}$

$= \dfrac{1}{\sqrt{49}+7}$

$= \dfrac{1}{7+7} = \boxed{\dfrac{1}{14}}$

④ $\lim\limits_{x\to -1} \dfrac{\sqrt{x+5}-2}{x+1} = \lim\limits_{x\to -1}\left[\dfrac{(\sqrt{x+5}-2)}{(x+1)}\dfrac{(\sqrt{x+5}+2)}{(\sqrt{x+5}+2)}\right]$

$= \lim\limits_{x\to -1}\left[\dfrac{(\sqrt{x+5})^2+2\sqrt{x+5}-2\sqrt{x+5}-4}{(x+1)(\sqrt{x+5}+2)}\right]$

$= \lim\limits_{x\to -1}\left[\dfrac{x+5-4}{(x+1)(\sqrt{x+5}+2)}\right]$

$= \lim\limits_{x\to -1}\left[\dfrac{x+1}{(x+1)(\sqrt{x+5}+2)}\right]$

$= \lim\limits_{x\to -1}\dfrac{1}{\sqrt{x+5}+2}$

$= \dfrac{1}{\sqrt{-1+5}+2}$

$= \dfrac{1}{\sqrt{4}+2}$

$= \dfrac{1}{2+2} = \boxed{\dfrac{1}{4}}$

⑤ $$\lim_{x\to 0}\frac{x}{\sqrt{16-x}-4}=\lim_{x\to 0}\left[\frac{x}{(\sqrt{16-x}-4)}\cdot\frac{(\sqrt{16-x}+4)}{(\sqrt{16-x}+4)}\right]$$

$$=\lim_{x\to 0}\left[\frac{x(\sqrt{16-x}+4)}{(\sqrt{16-x})^2+4\sqrt{16-x}-4\sqrt{16-x}-16}\right]$$

$$=\lim_{x\to 0}\left[\frac{x(\sqrt{16-x}+4)}{16-x-16}\right]$$

$$=\lim_{x\to 0}\left[\frac{x(\sqrt{16-x}+4)}{-x}\right]$$

$$=\lim_{x\to 0}\frac{\sqrt{16-x}+4}{-1}$$

$$=\frac{\sqrt{16-0}+4}{-1}$$

$$=\frac{\sqrt{16}+4}{-1}$$

$$=\frac{4+4}{-1}=\boxed{-8}$$

⑥ $$\lim_{x\to 36}\frac{\sqrt{x}-6}{x-36}=\lim_{x\to 36}\left[\frac{(\sqrt{x}-6)(\sqrt{x}+6)}{(x-36)(\sqrt{x}+6)}\right]$$

$$=\lim_{x\to 36}\left[\frac{(\sqrt{x})^2+6\sqrt{x}-6\sqrt{x}-36}{(x-36)(\sqrt{x}+6)}\right]$$

$$=\lim_{x\to 36}\left[\frac{x-36}{(x-36)(\sqrt{x}+6)}\right]$$

$$=\lim_{x\to 36}\frac{1}{\sqrt{x}+6}$$

$$=\frac{1}{\sqrt{36}+6}$$

$$=\frac{1}{6+6}$$

$$=\boxed{\frac{1}{12}}$$

Finding Limits by Simplifying Complex Fractions

7) $$\lim_{x\to 3}\frac{\frac{1}{x}-\frac{1}{3}}{x-3}=\lim_{x\to 3}\frac{\frac{1\cdot 3}{x\cdot 3}-\frac{1\cdot x}{3\cdot x}}{x-3}$$

$$=\lim_{x\to 3}\frac{\frac{3}{3x}-\frac{x}{3x}}{x-3}$$

$$=\lim_{x\to 3}\frac{\frac{3-x}{3x}}{x-3}$$

$$=\lim_{x\to 3}\left[\frac{3-x}{3x}\div(x-3)\right]$$

$$=\lim_{x\to 3}\left[\frac{3-x}{3x}\div\frac{x-3}{1}\right]$$

$$=\lim_{x\to 3}\left[\frac{3-x}{3x}\cdot\frac{1}{x-3}\right]$$

$$=\lim_{x\to 3}\left[\frac{3-x}{3x(x-3)}\right]$$

$$=\lim_{x\to 3}\left[\frac{-(-3+x)}{3x(x-3)}\right]$$

$$=\lim_{x\to 3}\frac{-1}{3x}$$

$$=\frac{-1}{3(3)}$$

$$=\boxed{-\frac{1}{9}}$$

8) $$\lim_{x\to -5}\frac{\frac{1}{7}+\frac{1}{x-2}}{2-\frac{8}{x+9}}=\lim_{x\to -5}\frac{\frac{1(x-2)}{7(x-2)}+\frac{1(7)}{(x-2)(7)}}{\frac{2(x+9)}{1(x+9)}-\frac{8}{x+9}}$$

$$=\lim_{x\to -5}\frac{\frac{x-2}{7(x-2)}+\frac{7}{7(x-2)}}{\frac{2x+18}{x+9}-\frac{8}{x+9}}$$

$$=\lim_{x\to -5}\frac{\frac{x+5}{7(x-2)}}{\frac{2x+10}{x+9}}$$

$$=\lim_{x\to -5}\left[\frac{x+5}{7(x-2)}\div\frac{2x+10}{x+9}\right]$$

$$=\lim_{x\to -5}\left[\frac{x+5}{7(x-2)}\cdot\frac{x+9}{2x+10}\right]$$

$$=\lim_{x\to -5}\left[\frac{(x+5)(x+9)}{7(x-2)(2x+10)}\right]$$

$$=\lim_{x\to -5}\left[\frac{(x+5)(x+9)}{14(x-2)(x+5)}\right]$$

$$=\lim_{x\to -5}\left[\frac{x+9}{14(x-2)}\right]$$

$$=\frac{-5+9}{14(-5-2)}$$

$$=\frac{4}{14(-7)}$$

$$=\boxed{-\frac{2}{49}}$$

9) $\lim_{x\to 0} \frac{-3x}{\frac{1}{9+x}-\frac{1}{9}} = \lim_{x\to 0} \frac{-3x}{\frac{1(9)}{(9+x)(9)}-\frac{1(9+x)}{9(9+x)}} = \lim_{x\to 0} \frac{-3x}{\frac{9}{9(9+x)}-\frac{9+x}{9(9+x)}}$

$= \lim_{x\to 0} \frac{-3x}{\frac{9-(9+x)}{9(9+x)}}$

$= \lim_{x\to 0} \frac{-3x}{\frac{9-9-x}{9(9+x)}}$

$= \lim_{x\to 0} \frac{-3x}{\frac{-x}{9(9+x)}}$

$= \lim_{x\to 0} \left[-3x \div \frac{-x}{9(9+x)}\right]$

$= \lim_{x\to 0} \left[\frac{-3x}{1} \cdot \frac{9(9+x)}{-x}\right]$

$= \lim_{x\to 0} \frac{-27x(9+x)}{-x}$

$= \lim_{x\to 0} 27(9+x)$

$= 27(9+0)$

$= \boxed{243}$

10) $\lim_{x\to -4} \frac{\frac{1}{4}+\frac{1}{x}}{x+4} = \lim_{x\to -4} \frac{\frac{1\cdot x}{4\cdot x}+\frac{1\cdot 4}{x\cdot 4}}{x+4} = \lim_{x\to -4} \frac{\frac{x}{4x}+\frac{4}{4x}}{x+4}$

$= \lim_{x\to -4} \frac{\frac{x+4}{4x}}{x+4}$

$= \lim_{x\to -4} \left[\frac{x+4}{4x} \div (x+4)\right]$

$= \lim_{x\to -4} \left[\frac{x+4}{4x} \div \frac{x+4}{1}\right]$

$= \lim_{x\to -4} \left[\frac{x+4}{4x} \cdot \frac{1}{x+4}\right]$

$= \lim_{x\to -4} \left[\frac{x+4}{4x(x+4)}\right]$

$= \lim_{x\to -4} \frac{1}{4x}$

$= \frac{1}{4(-4)}$

$= \boxed{-1/16}$

(11)
$$\lim_{x\to0}\frac{\frac{1}{x+2}-\frac{1}{2}}{x}=\lim_{x\to0}\frac{\frac{1}{(x+2)}\frac{(2)}{(2)}-\frac{1}{2}\frac{(x+2)}{(x+2)}}{x}$$

$$=\lim_{x\to0}\frac{\frac{2}{2(x+2)}-\frac{x+2}{2(x+2)}}{x}$$

$$=\lim_{x\to0}\frac{\frac{2-(x+2)}{2(x+2)}}{x}$$

$$=\lim_{x\to0}\frac{\frac{2-x-2}{2(x+2)}}{x}$$

$$=\lim_{x\to0}\frac{\frac{-x}{2(x+2)}}{x}$$

$$=\lim_{x\to0}\left[\frac{-x}{2(x+2)}\div x\right]$$

$$=\lim_{x\to0}\left[\frac{-x}{2(x+2)}\div\frac{x}{1}\right]$$

$$=\lim_{x\to0}\left[\frac{-x}{2(x+2)}\cdot\frac{1}{x}\right]$$

$$=\lim_{x\to0}\left[\frac{-x}{2x(x+2)}\right]$$

$$=\lim_{x\to0}\frac{-1}{2(x+2)}$$

$$=\frac{-1}{2(0+2)}$$

$$=\boxed{-\frac{1}{4}}$$

(12)
$$\lim_{x\to2}\frac{\frac{6}{x}-3}{\frac{1}{6}-\frac{1}{x+4}}=\lim_{x\to2}\frac{\frac{6}{x}-\frac{3\cdot x}{1\cdot x}}{\frac{1(x+4)}{6(x+4)}-\frac{1(6)}{(x+4)(6)}}$$

$$=\lim_{x\to2}\frac{\frac{6}{x}-\frac{3x}{x}}{\frac{x+4}{6(x+4)}-\frac{6}{6(x-4)}}$$

$$=\lim_{x\to2}\frac{\frac{6-3x}{x}}{\frac{x-2}{6(x+4)}}$$

$$=\lim_{x\to2}\left[\frac{6-3x}{x}\div\frac{x-2}{6(x+4)}\right]$$

$$=\lim_{x\to2}\left[\frac{6-3x}{x}\cdot\frac{6(x+4)}{x-2}\right]$$

$$=\lim_{x\to2}\left[\frac{6(6-3x)(x+4)}{x(x-2)}\right]$$

$$=\lim_{x\to2}\left[\frac{(6)(-3)(-2+x)(x+4)}{x(x-2)}\right]$$

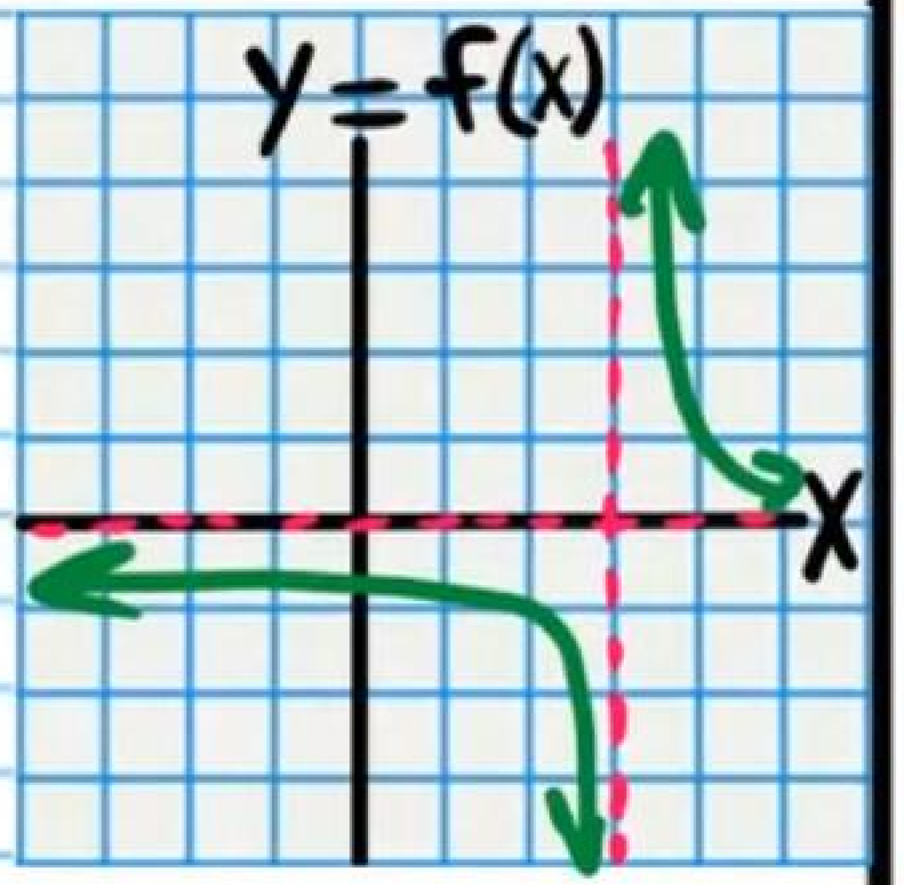
$$=\lim_{x\to2}\left[\frac{-18(x+4)}{x}\right]$$

$$=\frac{-18(2+4)}{2}$$

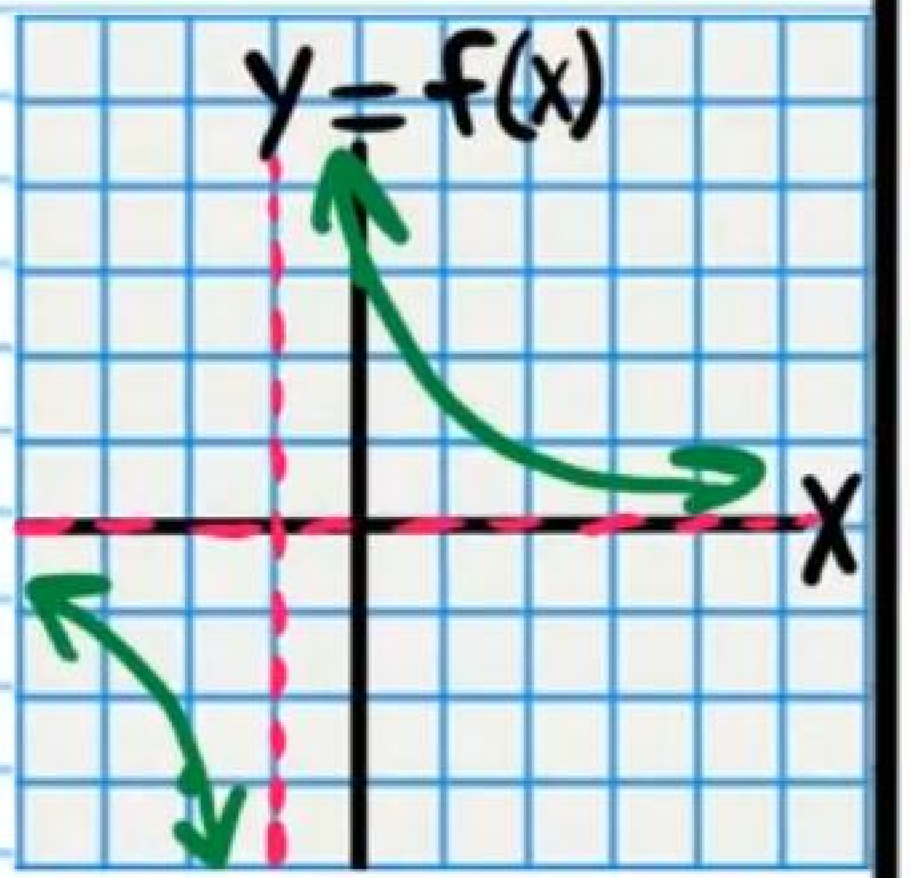
$$=\boxed{-54}$$

Finding Limits at Infinite Discontinuities

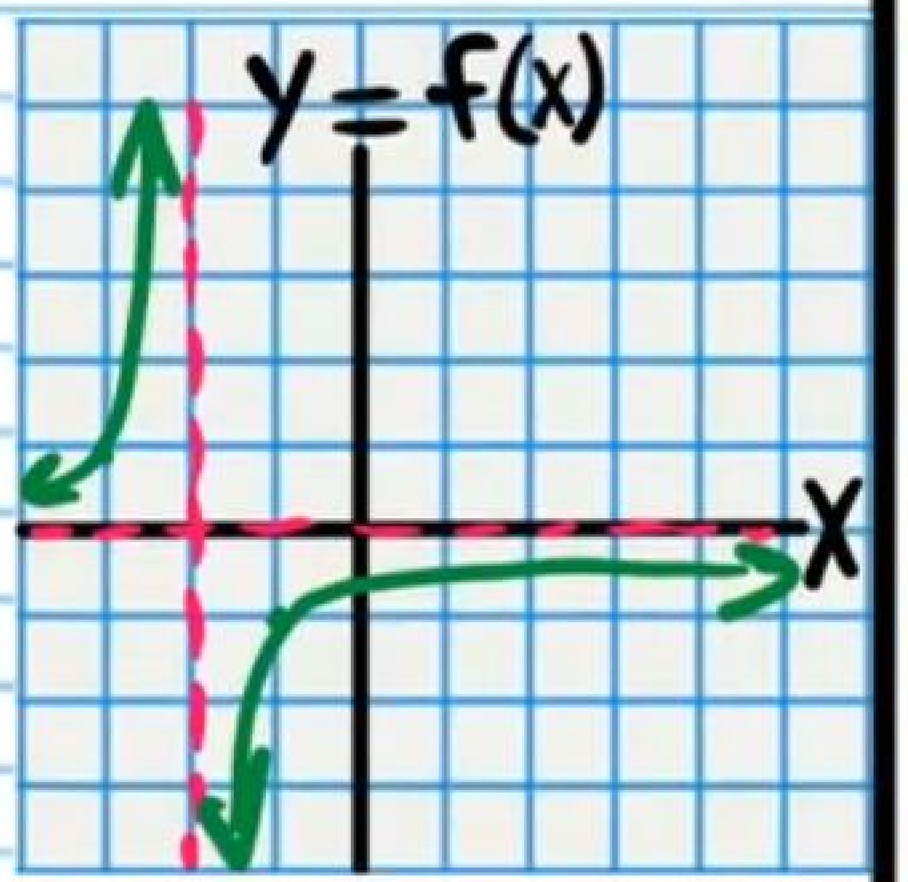
(13) $\lim_{x \to 3^+} \frac{1}{x-3} = \boxed{\infty}$



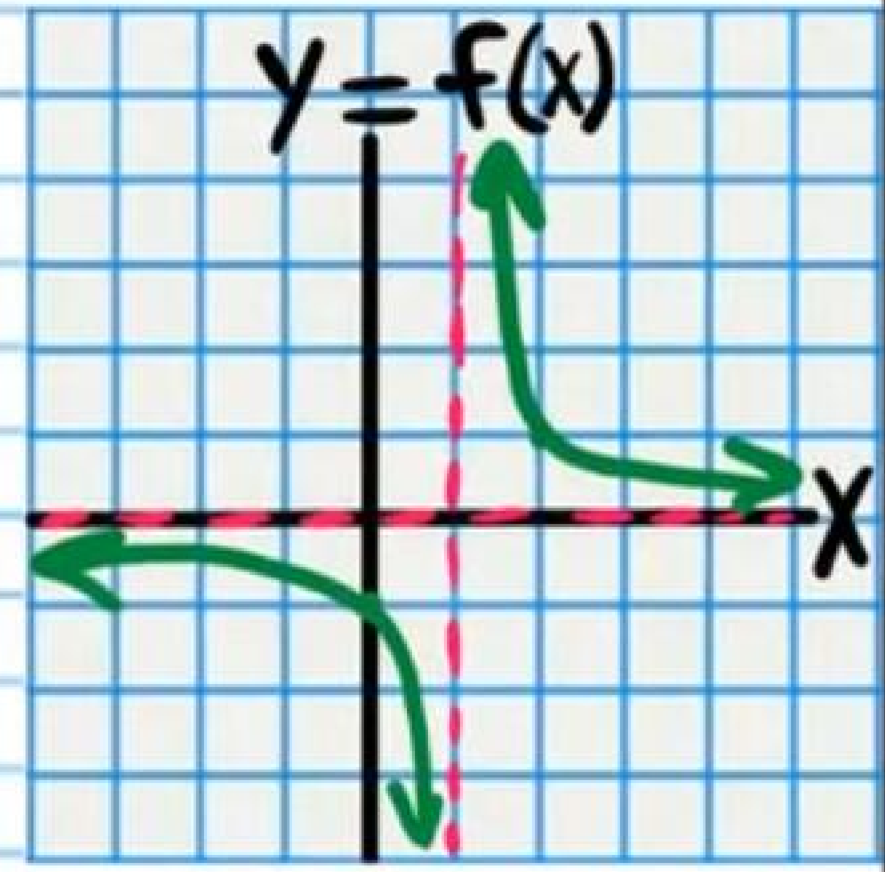
(14) $\lim_{x \to -1^-} \frac{3}{x+1} = \boxed{-\infty}$



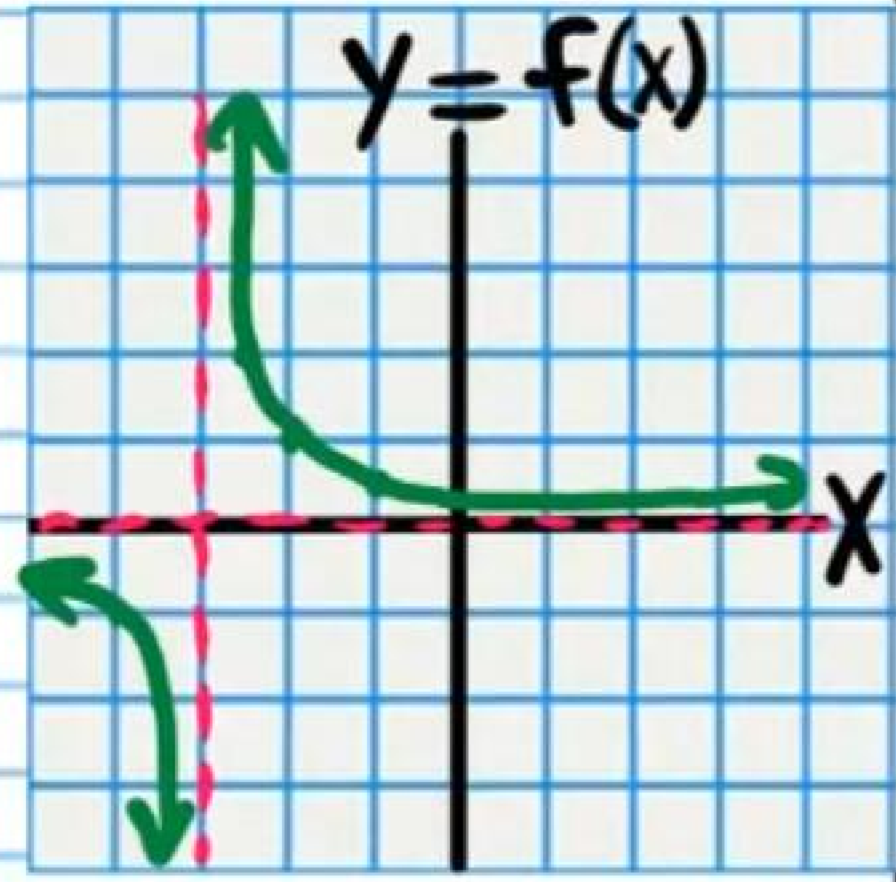
(15) $\lim_{x \to -2^+} \frac{-1}{x+2} = \boxed{-\infty}$



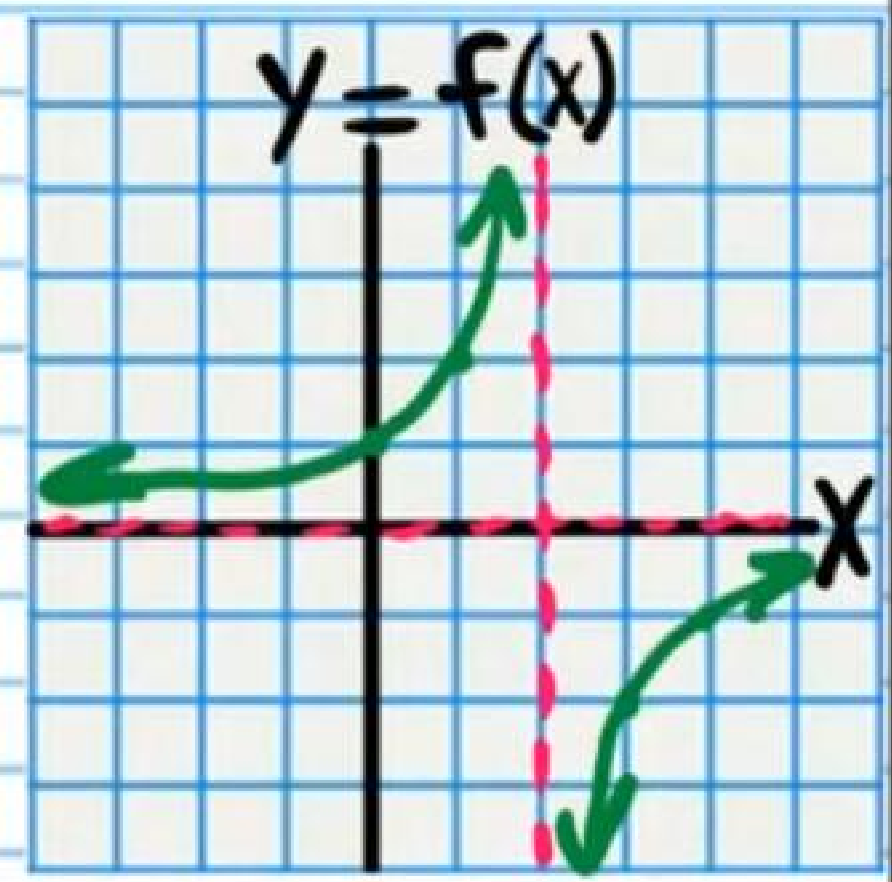
(16) $\lim_{x \to 1^-} \frac{1}{x-1} = \boxed{-\infty}$



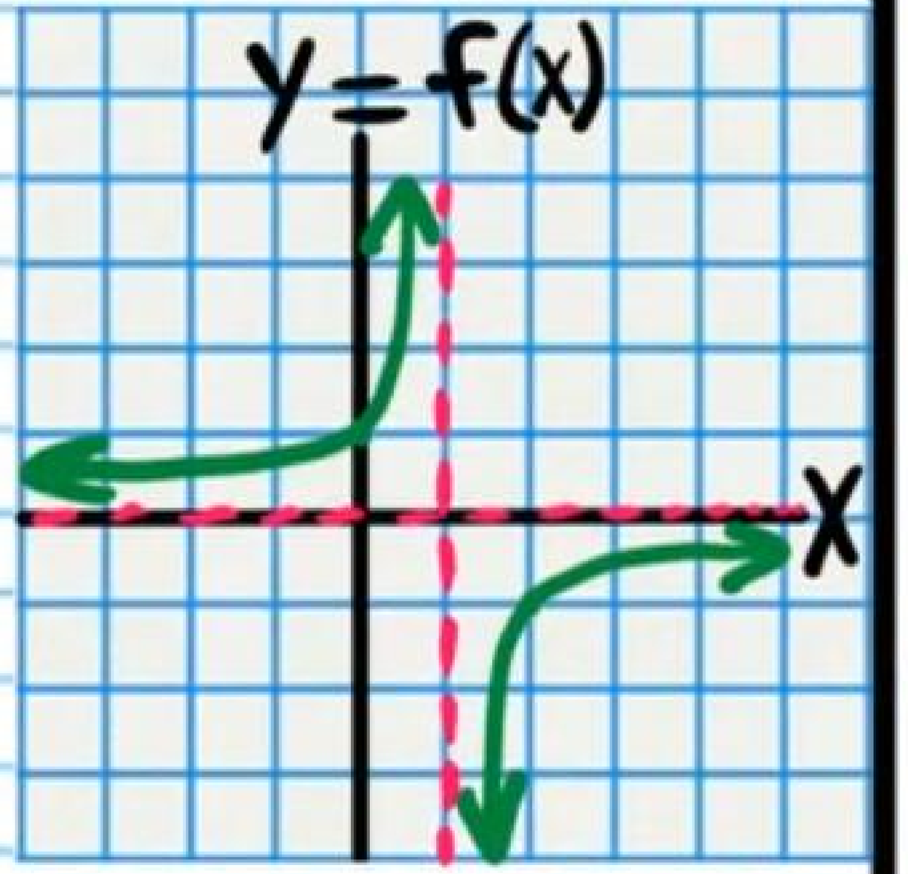
(17) $\lim_{x \to -3^+} \frac{1}{x+3} = \boxed{\infty}$



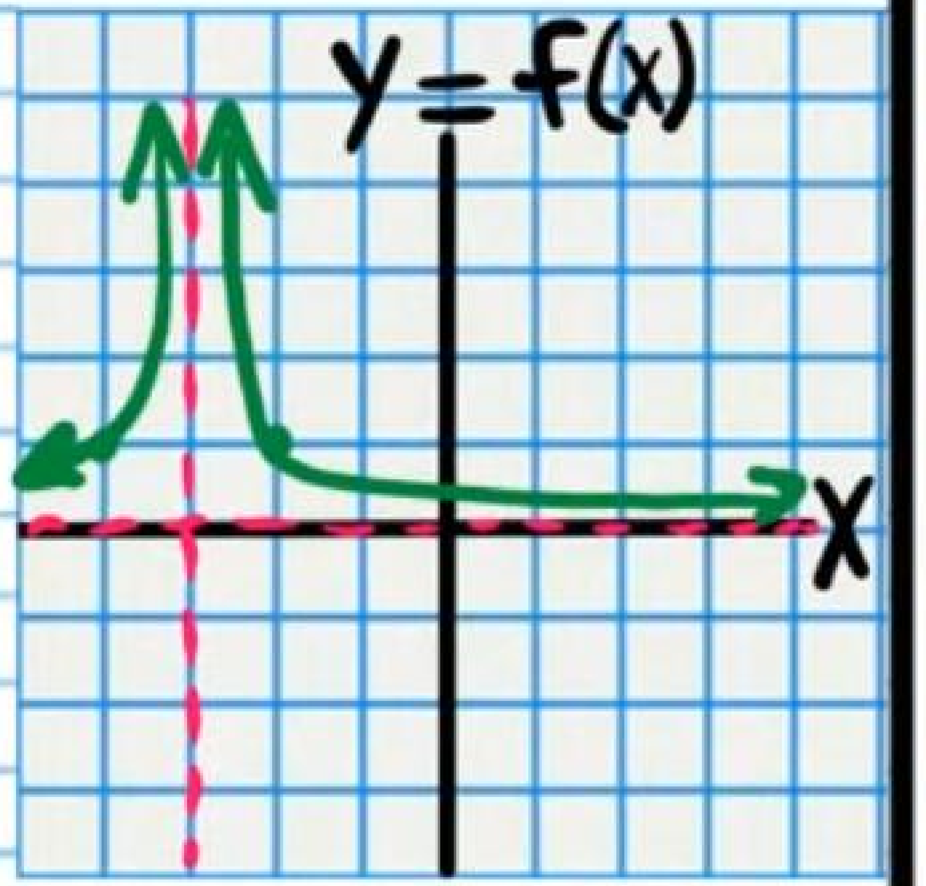
(18) $\lim_{x \to 2^-} \frac{-2}{x-2} = \boxed{\infty}$



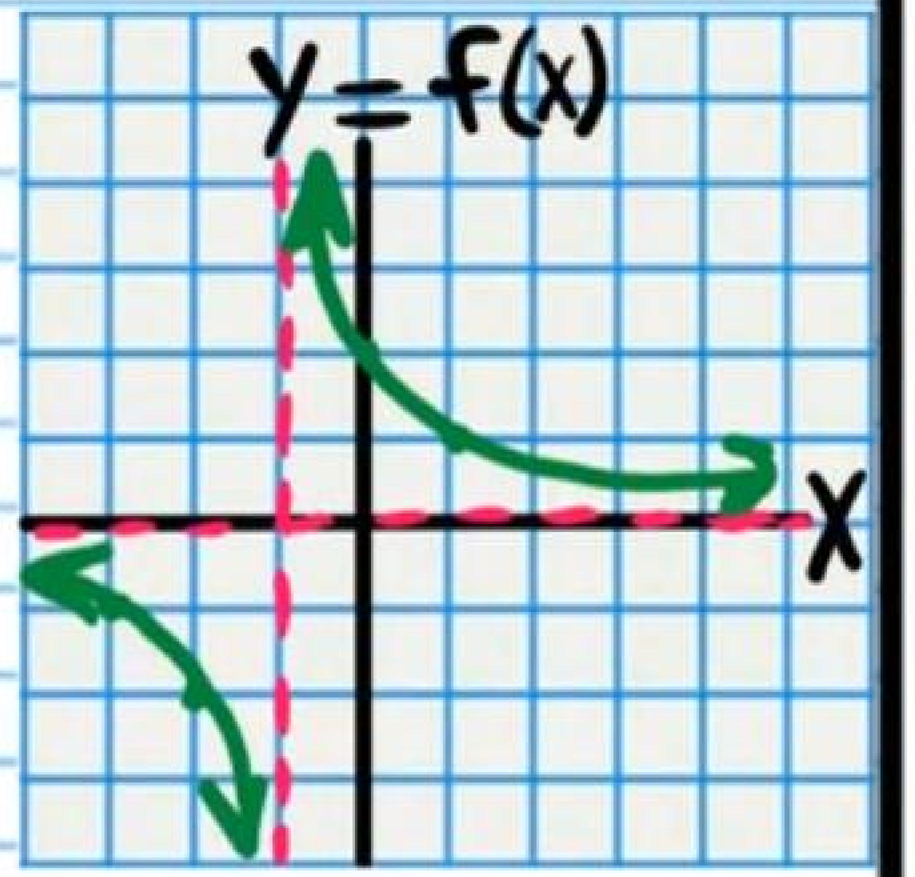
(19) $\lim\limits_{x\to 1} \frac{-1}{x-1} = \boxed{DNE}$



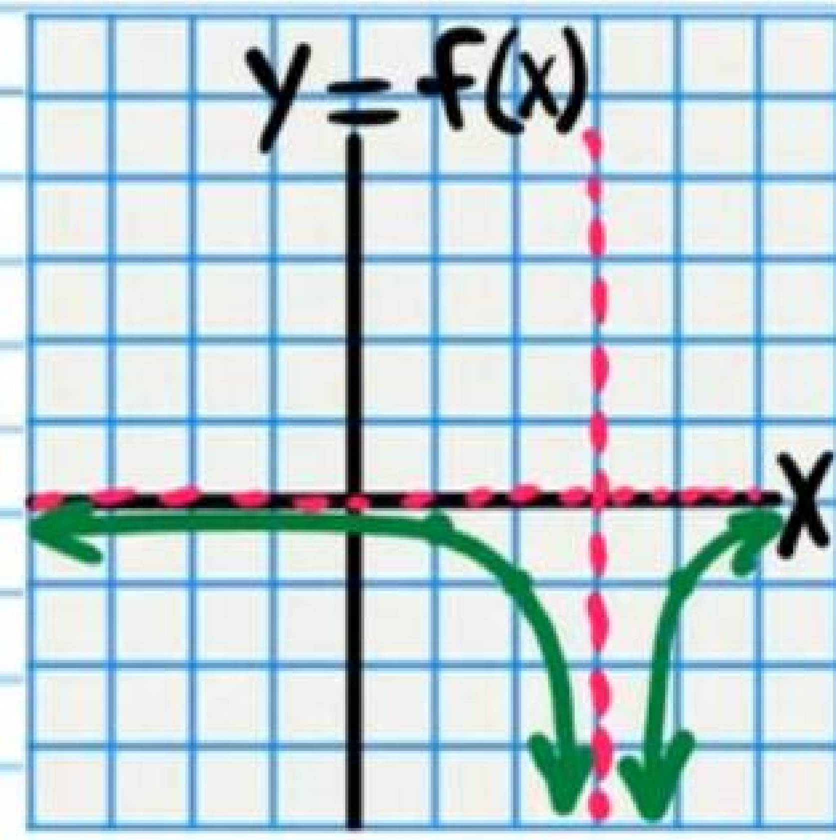
(20) $\lim\limits_{x\to -3} \frac{1}{(x+3)^2} = \boxed{\infty}$



(21) $\lim\limits_{x\to -1} \frac{2}{x+1} = \boxed{DNE}$

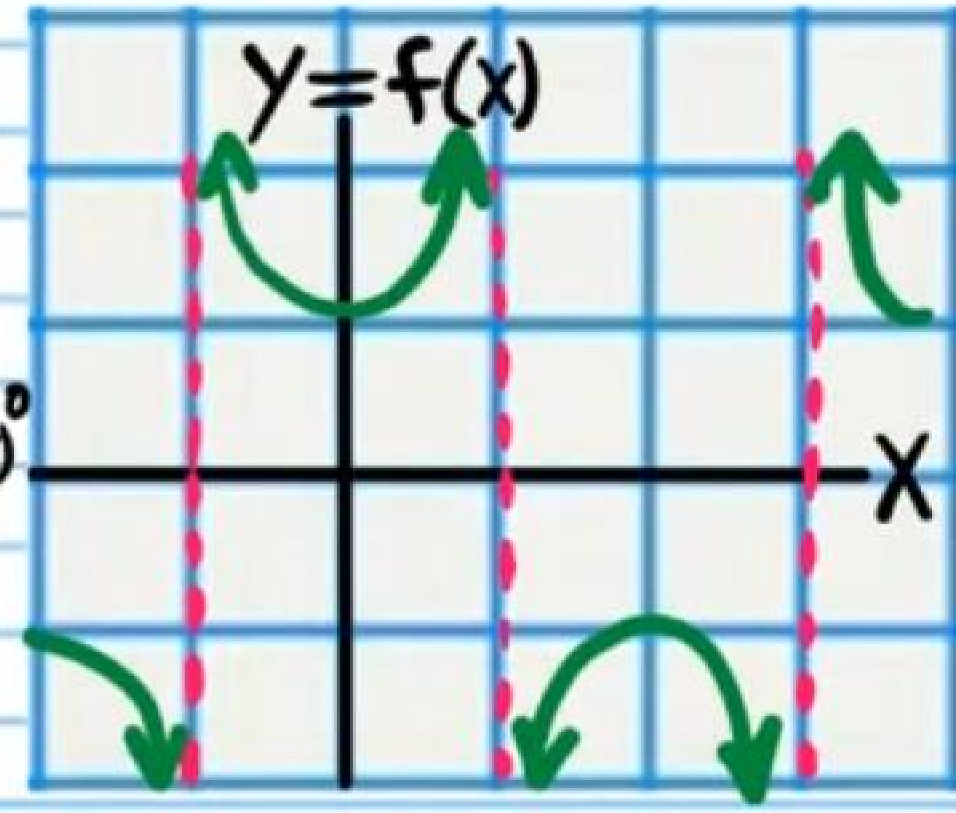


(22) $\lim\limits_{x\to 3} \frac{-1}{(x-3)^2} = \boxed{-\infty}$

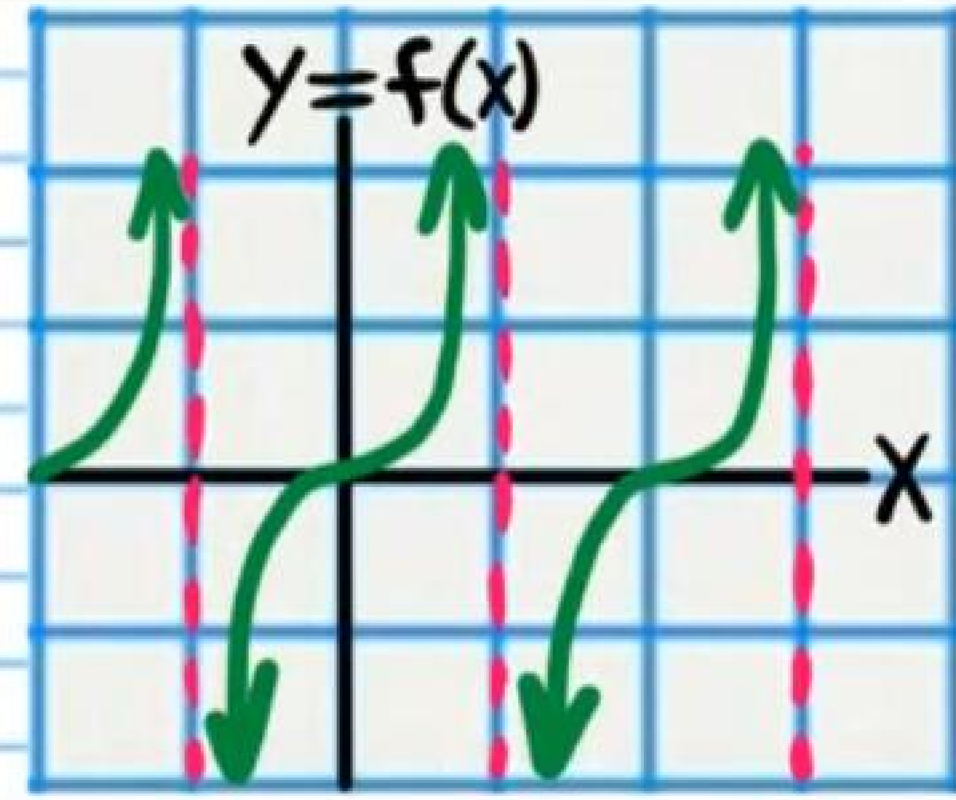


(23) $\lim\limits_{x\to \frac{\pi}{2}^+} \sec x = \boxed{-\infty}$

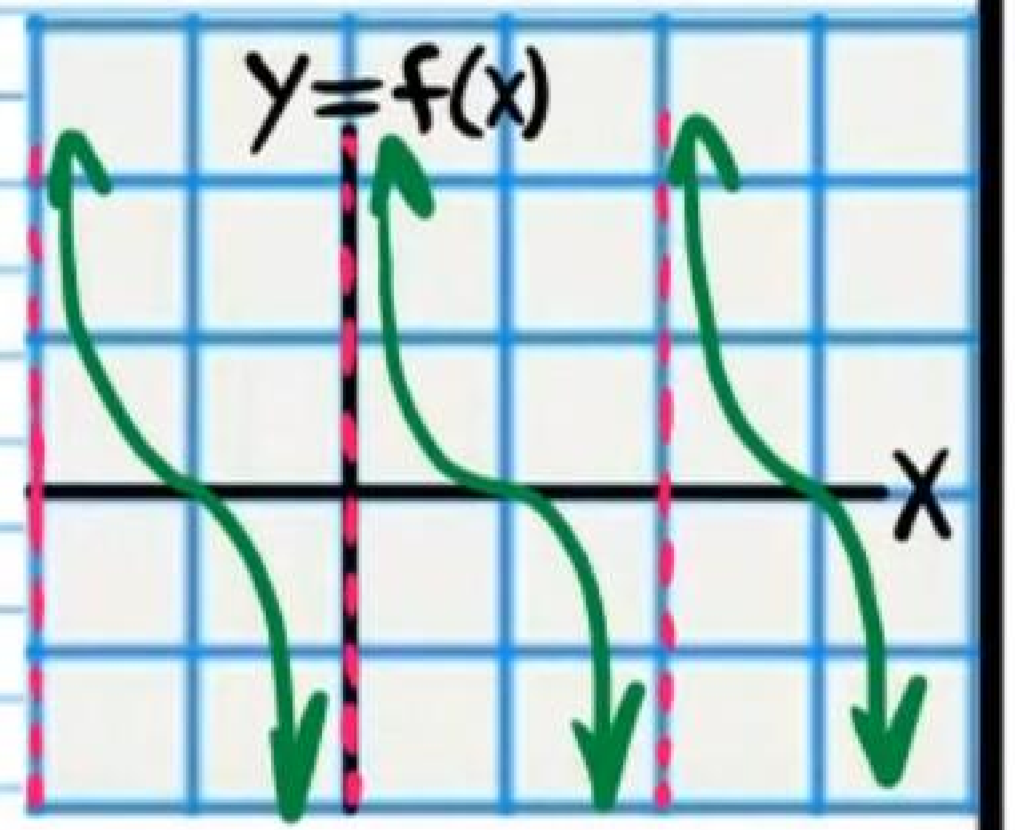
$\frac{\pi}{2} \times \frac{180^\circ}{\pi} = \frac{180^\circ}{2} = 90^\circ$



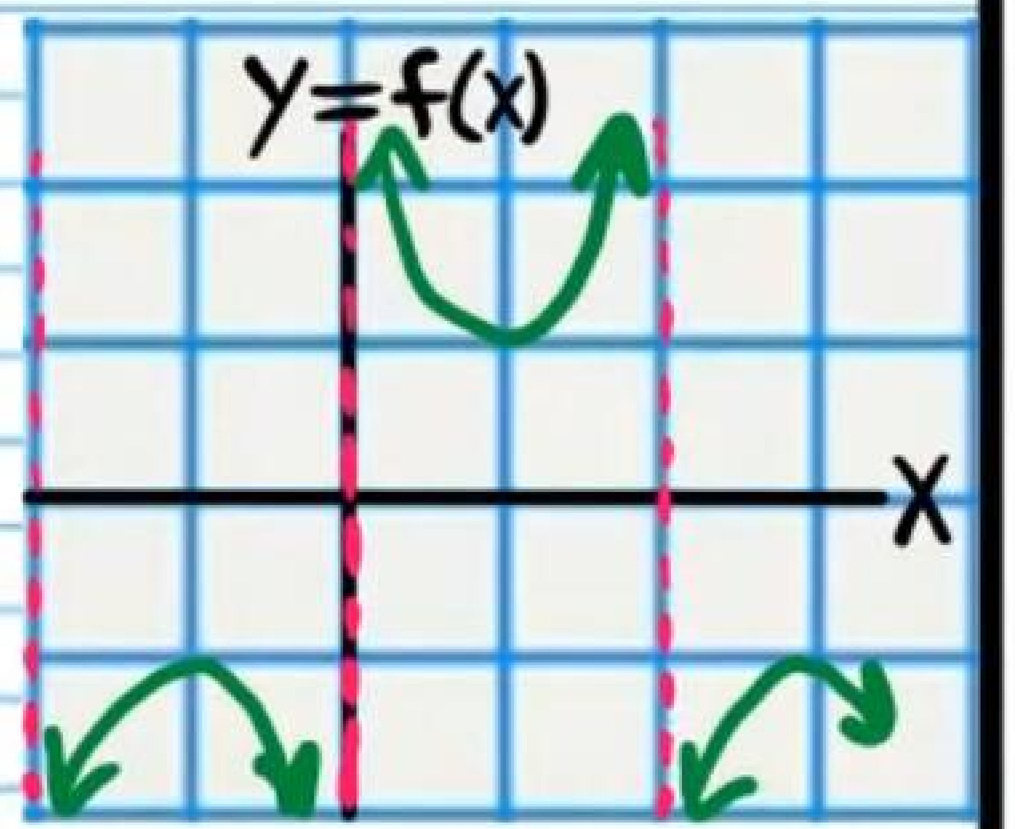
(24) $\lim\limits_{x\to -\frac{\pi}{2}^-} \tan x = \boxed{\infty}$



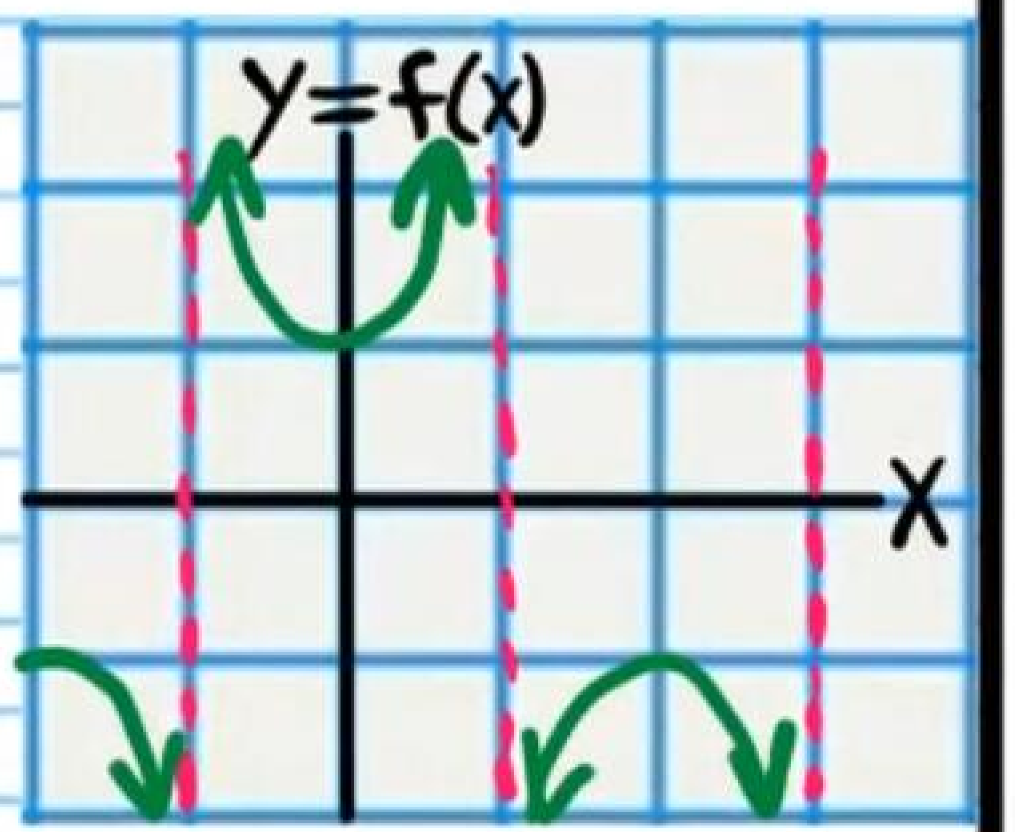
(25) $\lim_{x \to \pi} \cot x = \boxed{DNE}$



(26) $\lim_{x \to 0^-} \csc x = \boxed{-\infty}$



(27) $\lim_{x \to -\frac{\pi}{2}} \sec x = \boxed{DNE}$



(28) $\lim_{x \to \pi^+} \cot x = \boxed{\infty}$